CAMSHAFT POSITION SENSOR - 2.4L (Continued)

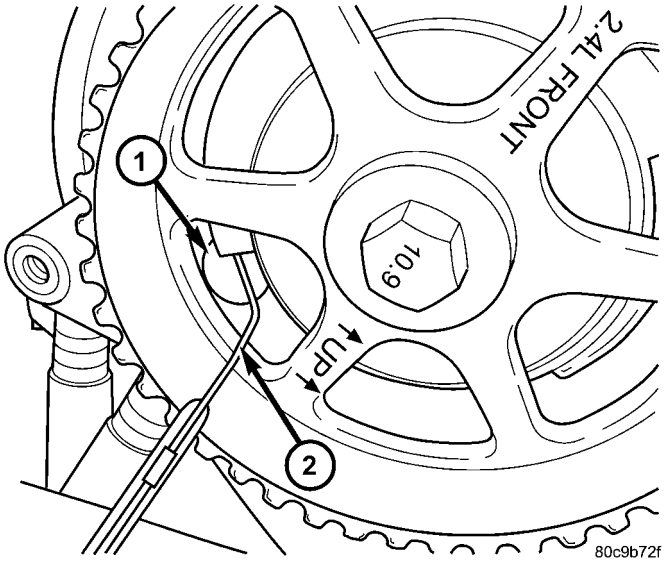

Fig. 10 CMP ADJUSTMENT - 2.4L

1 - FACE OF SENSOR
2 - WIRE GAPPING TOOL

CAMSHAFT POSITION SENSOR - 4.0L

DESCRIPTION - 4.0L

The CMP (Camshaft Position Sensor) sensor (2) (Fig. 11) is bolted to the side of the oil pump drive assembly (5). The assembly is located on the right side of the engine near the oil filter (Fig. 11).

OPERATION - 4.0L

The CMP sensor (1) (Fig. 12) uses a rotating hall effect device called a target wheel (3) which is attached to the oil pump drive shaft. The target wheel contains sets of machined notches (2).

When the leading edge of a target wheel notch passes by the tip of the CMP sensor, the interruption of magnetic field causes the voltage to switch high resulting in a sync signal of approximately 5 volts.

When the trailing edge of the target wheel notch passes by the tip of the CMP sensor, the change of the magnetic field causes the sync signal voltage to switch low to 0 volts.

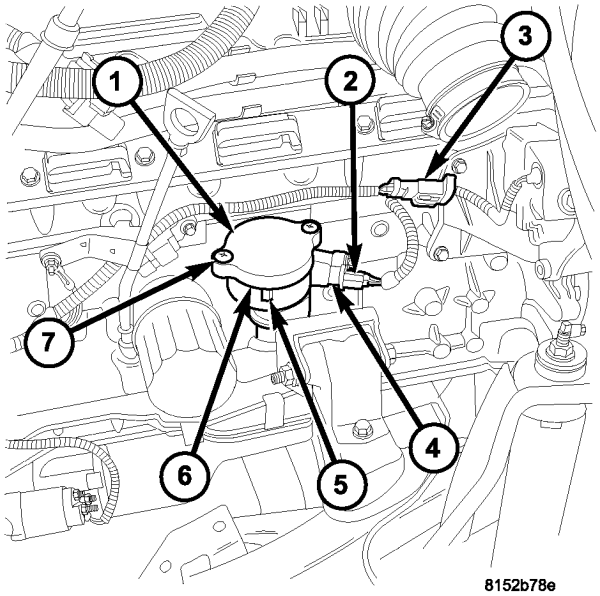

Fig. 11 4.0L CMP LOCATION ('05)

1 - OIL PUMP DRIVE ASSEMBLY
2 - CMP
3 - CMP ELECTRICAL CONNECTOR
4 - HOUSING HOLD DOWN BOLT
5 - HOUSING
6 - PLASTIC COVER
7 - COVER SCREWS (2)

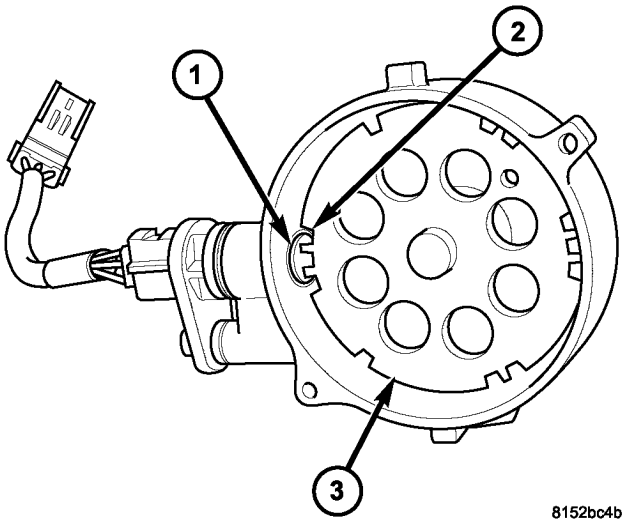

Fig. 12 4.0L CMP OPERATION

1 - CMP SENSOR
2 - TARGET WHEEL NOTCHES
3 - TARGET WHEEL